ANKRD26-Related Thrombocytopenia (THC2): A Comprehensive Disease Characteristics Report

Disease: ANKRD26-Related Thrombocytopenia (THC2) **MONDO ID:** MONDO:0008555 | **OMIM:** 188000 (THC2 phenotype; *ANKRD26* gene MIM 610855) | **Causal gene:** *ANKRD26* (Ankyrin Repeat Domain 26; HGNC:21026; NCBI Gene 22852; chromosome 10p12.1) | **Orphanet:** ORPHA:3322 | **Synonyms:** THC2, Thrombocytopenia 2, ANKRD26-RT **Category:** Mendelian, autosomal dominant

Summary

ANKRD26-Related Thrombocytopenia (THC2) is an autosomal-dominant inherited platelet-production disorder caused by germline gain-of-expression point variants (and small indels) in the 5′ untranslated region (5′UTR) of the *ANKRD26* gene. Despite the serial "thrombocytopenia 2" name, the causal gene is unambiguously *ANKRD26* (not the neighboring *MASTL* or *ACBD5* genes at the same 10p12.1 locus, which were historically proposed and later excluded by fine-mapping). The disease is defined by lifelong, usually mild-to-moderate thrombocytopenia with a hallmark distinguishing feature — **normal platelet size (normocytic platelets)** — which separates it from most other inherited thrombocytopenias that present with macrothrombocytopenia.

The core molecular lesion is loss of transcriptional repression. Recurrent single-nucleotide variants clustered in a highly conserved 19-bp region of the *ANKRD26* 5′UTR abolish binding of the master megakaryopoietic repressors **RUNX1 and FLI1**. Normally these factors silence *ANKRD26* during terminal megakaryocyte (MK) maturation; when their binding is lost, *ANKRD26* expression persists inappropriately. Sustained ANKRD26 protein stabilizes the type I cytokine receptors MPL, EPOR and G-CSFR by preventing their internalization, hyperactivates the **MAPK/ERK** signaling cascade, and impairs proplatelet formation. Single-cell studies further show that elevated ANKRD26 localizes to the centrosome and drives **JUNB-mediated transcriptional activation of p21**, inducing apoptosis of polyploid megakaryocytes — producing the dysmegakaryopoiesis that underlies the low platelet count.

The principal clinical significance of THC2 is not its usually-mild bleeding tendency but its **substantial lifetime predisposition to myeloid malignancy** (AML, MDS, CMML; estimated 10–45% across the RUNX1/ETV6/ANKRD26 group of inherited-thrombocytopenia-with-malignancy-risk disorders). There is no disease-modifying therapy. Management is supportive and centers on avoiding misdiagnosis as immune thrombocytopenia (ITP), perioperative platelet support (with the TPO-receptor agonist eltrombopag as an effective alternative to transfusion), genetic counseling with cascade testing, informed donor selection for any future hematopoietic stem cell transplant, and lifelong hematologic surveillance for clonal evolution.

> **Evidence basis:** This report integrates human clinical cohorts/registries, patient-derived primary megakaryocytes and iPSCs, single-cell bone-marrow profiling, and one murine knockout. No proprietary patient-level/EHR data were used; all content derives from aggregated disease-level literature and ontologies.

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1. Disease Information

Overview. THC2 is a Mendelian, autosomal-dominant, non-syndromic inherited thrombocytopenia caused by germline variants in the 5′UTR of *ANKRD26*. It is characterized by mild-to-moderate reduction in platelet count, normal-sized platelets, usually mild (or absent) bleeding, dysmegakaryopoiesis in the bone marrow, and a clinically important predisposition to hematologic (predominantly myeloid) neoplasms.

Key identifiers.

Resource	Identifier
MONDO	MONDO:0008555
OMIM	188000 (THC2 phenotype); *ANKRD26* gene MIM 610855
Orphanet	ORPHA:3322
Gene (HGNC)	*ANKRD26*, HGNC:21026
NCBI Gene	22852
Gene locus	10p12.1
Synonyms	THC2, Thrombocytopenia 2, ANKRD26-related thrombocytopenia (ANKRD26-RT)

Information source. The knowledge in this report derives predominantly from **aggregated disease-level resources** — cohort studies, inherited-thrombocytopenia registries (Italian and Japanese), case series, and functional laboratory studies — rather than from individual EHR records. The largest clinical description is a cohort of 78 patients from 21 families PMID: 21467542.

2. Etiology

Disease causal factors. THC2 is a purely **genetic (Mendelian)** disorder. There is no known environmental, infectious, or acquired cause of the primary disease. The causal lesion is a germline point mutation (or small deletion) in the conserved 19-bp region of the *ANKRD26* 5′UTR that disrupts RUNX1/FLI1 repressor binding, producing a **gain of ANKRD26 expression** (not haploinsufficiency) PMID: 21211618.

Genetic risk factors. The causal variants themselves are the primary genetic risk factor. Six distinct 5′UTR mutations were originally identified across 8 unrelated families, all clustered in the conserved 19-bp hotspot and absent from 500 controls and the 1000 Genomes database PMID: 21211618. Recurrent hotspot positions span roughly nucleotides −140 to −113 (e.g., c.-128G>A, c.-127A>T, c.-126T>G, c.-125T>G, c.-118C>T, c.-116C>T, c.-113A>C), plus later-described variants such as the deletion c.-129_-112del that disrupts the RUNX1-binding site PMID: 40954090.

Modifier genes. *GATA2* acts as a **positive regulator** of *ANKRD26* transcription in myeloid-derived cells, opposing RUNX1/FLI1 repression and adding a regulatory layer to the GATA2–ANKRD26 axis PMID: 38817334. Somatic driver mutations acquired in hematopoietic regulators on the germline *ANKRD26* background are the key modifiers of malignant transformation risk (see §4, §11).

Environmental / lifestyle / protective factors. No environmental risk factors, protective variants, or gene–environment interactions have been established for THC2. This is consistent with a monogenic germline disorder. *Not applicable / not available* for most environmental subcategories.

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3. Phenotypes

THC2 phenotypes are predominantly **laboratory abnormalities** and **hemorrhagic clinical signs**, usually mild.

Phenotype	Type	HPO suggestion	Onset	Severity	Frequency
Thrombocytopenia	Lab abnormality	HP:0001873	Congenital/ lifelong	Mild–moderate	~100% (defining)
Normal platelet size (normocytic)	Lab/ morphology	(normal MPV; absence of HP:0011877)	Lifelong	—	Nearly all patients
Bleeding tendency (mild; mucocutaneous, bruising)	Clinical sign	HP:0001892 (abnormal bleeding)	Variable	Mild	~25–30%
Glycoprotein Ia (ITGA2/GPIa) deficiency	Lab abnormality	HP:0011869 (platelet dysfunction)	Lifelong	—	Majority of cases
α-granule deficiency	Lab/ morphology	HP:0030401	Lifelong	—	Majority of cases
Dysmegakaryopoiesis (marrow)	Lab/ pathology	HP:0012757	Lifelong	—	Characteristic
Elevated serum thrombopoietin	Lab abnormality	—	Lifelong	—	Common
Predisposition to myeloid neoplasia	Clinical course	HP:0004377 (hematological neoplasm)	Adult (variable)	Severe when present	10–45% (group estimate)
Occasional high hemoglobin / leukocytosis	Lab abnormality	HP:0001899 / HP:0001974	Variable	Mild	A few cases

Characteristics. In the 78-patient / 21-family cohort, "thrombocytopenia and bleeding tendency were usually mild. Nearly all patients had no platelet macrocytosis, and this characteristic distinguishes THC2 from most other forms of inherited thrombocytopenia. In the majority of cases, platelets were deficient in glycoprotein Ia and α-granules, whereas in vitro platelet aggregation was normal" PMID: 21467542. A more recent multicenter cohort tempered the "normal aggregation" description: platelet aggregation was impaired in 90% of all

inherited-thrombocytopenia patients studied, with heterogeneous activation and granule secretion, and bleeding tendency present in 25–30% of ANKRD26-RT (and RUNX1-RT) patients — "Bleeding tendency was present in 25%-30% of RUNX1-RT and ANKRD26-RT patients. Platelet aggregation was impaired in 90% of all patients, while reduced activation and granule secretion were heterogeneous" PMID: 40670159.

Quality of life impact. Because bleeding is generally mild, the day-to-day QoL impact of the thrombocytopenia itself is modest for most patients. The dominant QoL burden is **psychological and surveillance-related** — living with a cancer-predisposition diagnosis and undergoing lifelong monitoring — and the morbidity of any malignancy that develops. Formal EQ-5D/SF-36 data specific to THC2 are *not available*.

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4. Genetic / Molecular Information

Causal gene. *ANKRD26* (Ankyrin Repeat Domain 26), 10p12.1, HGNC:21026, NCBI Gene 22852. *ANKRD26* is the evolutionary ancestor of the primate-specific *POTE* proapoptotic gene family PMID: 21211618.

Pathogenic variants. - **Variant type/class:** Predominantly single-nucleotide substitutions in the 5′UTR; also small deletions (e.g., c.-129_-112del). These are **regulatory (non-coding) variants**, an unusual class for a Mendelian disorder. - **Location:** Clustered in a highly conserved 19-bp sequence of the 5′UTR, spanning approximately nucleotides −140 to −113. - **Classification (ACMG/AMP):** The recurrent hotspot 5′UTR variants are classified **pathogenic/likely pathogenic** in ClinVar; novel 5′UTR variants outside the canonical hotspot and coding variants of uncertain significance (VUS) require functional interpretation. - **Allele frequency:** Extremely rare — original mutations absent from 500 controls and the 1000 Genomes database PMID: 21211618. - **Origin: Germline.** Somatic mutations in other genes accrue on this background during leukemic transformation. - **Functional consequence: Gain of expression** (not loss of function / not haploinsufficiency). "The luciferase reporter assay suggests that these 5′ UTR mutations might enhance *ANKRD26* expression," and "available data from an animal model and Dr. Watson's genome give evidence against haploinsufficiency as the pathogenetic mechanism" PMID: 21211618.

Mechanistic basis of the gain of expression. THC2 5′UTR mutations cause **loss of RUNX1 and FLI1 binding**: "THC2-associated mutations in the 5′ UTR of ANKRD26 resulted in loss of runt-related transcription factor 1 (RUNX1) and friend leukemia integration 1 transcription factor (FLI1) binding" PMID: 24430186. A Japanese registry independently confirmed a novel variant

that "disrupted the runt-related transcription factor 1 (RUNX1)-binding site, leading to ANKRD26 overexpression and possible haematopoietic abnormalities," with higher transcript levels ($p<0.001$), reduced proplatelet formation, and lower nuclear ploidy PMID: 40954090.

Modifier genes. *GATA2* positively regulates *ANKRD26* PMID: 38817334. Somatic drivers in hematopoietic regulators determine transformation PMID: 35751752.

Epigenetic / chromosomal. No recurrent epigenetic signature or large chromosomal abnormality defines germline THC2. Clonal cytogenetic/molecular changes appear only with malignant transformation. *Not applicable* for the primary disorder.

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5. Environmental Information

No environmental factors, lifestyle factors, or infectious agents are known to cause or trigger THC2. It is a monogenic germline disorder. General hematologic prudence (avoiding unnecessary antiplatelet agents, and caution with leukemogenic exposures given the cancer predisposition) is advisable, but no specific environmental trigger is established. **Not applicable** for environmental etiology.

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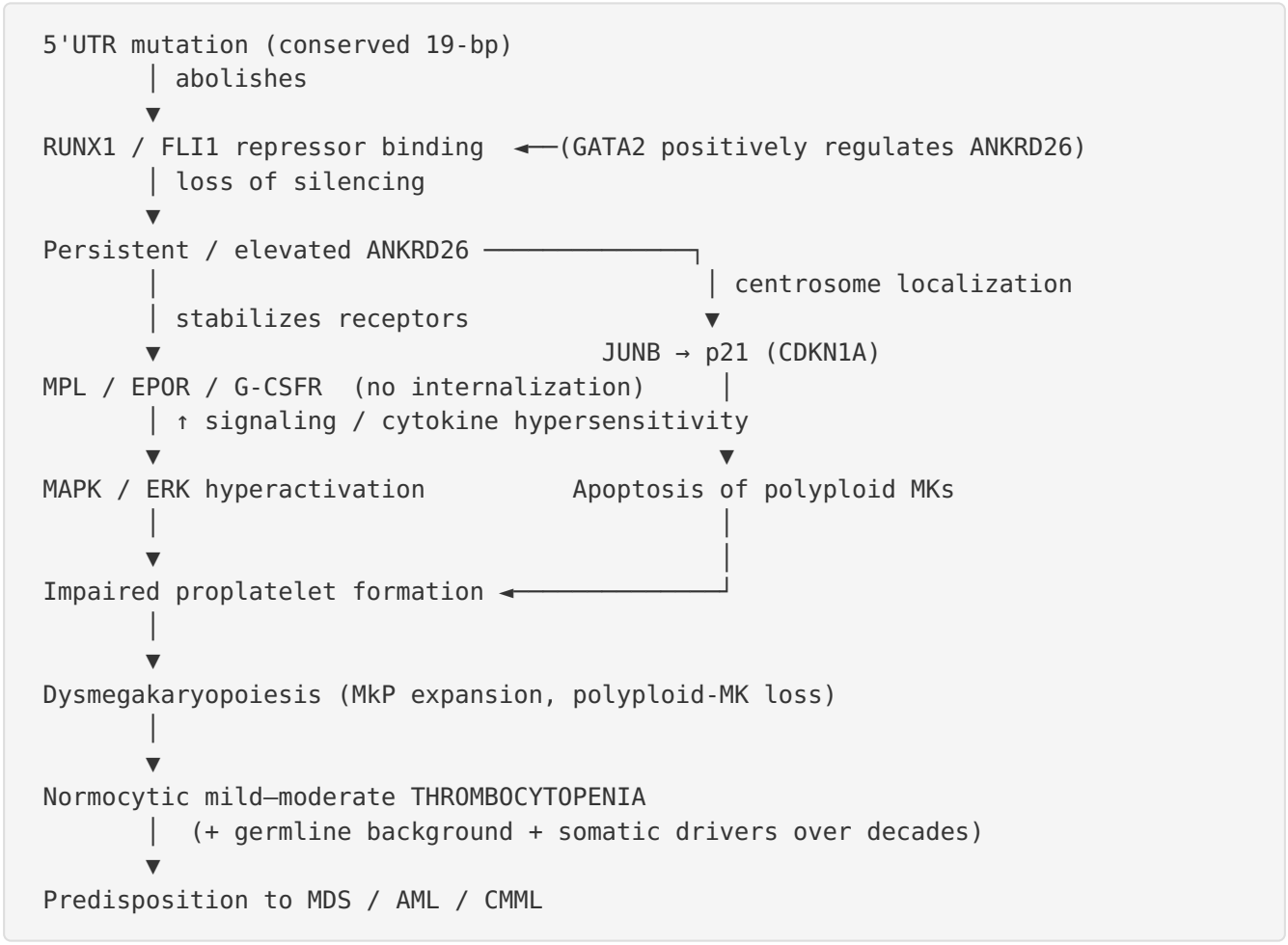
6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. A germline point mutation (or small deletion) arises in the conserved 19-bp region of the *ANKRD26* 5′UTR → **destroys the RUNX1/FLI1 repressor binding site** (demonstrated) [PMID: 24430186].
2. Loss of RUNX1/FLI1 binding → **failure to silence *ANKRD26* during terminal megakaryopoiesis** → sustained/elevated ANKRD26 protein (demonstrated by reporter assays and patient MKs) [PMID: 21211618, 24430186].
3. Elevated ANKRD26 **binds and stabilizes the type I cytokine receptors MPL, EPOR and G-CSFR by preventing their internalization** → increased receptor signaling and cytokine hypersensitivity (demonstrated) [PMID: 36794499].
4. Increased MPL/TPO signaling → **hyperactivation of the MAPK/ERK cascade** (demonstrated) [PMID: 24430186].

5. **Branch A — maturation/proplatelet defect:** MAPK/ERK hyperactivation → **impaired proplatelet formation** by megakaryocytes; ERK inhibition completely rescues proplatelet formation in vitro, proving causality [PMID: 24430186].
6. **Branch B — apoptotic depletion of mature MKs:** elevated ANKRD26 localizes to the **centrosome** (implicating mitotic regulation) → drives **JUNB-mediated transcriptional activation of p21 → apoptosis of polyploid megakaryocytes** (demonstrated by single-cell profiling) [PMID: 41538704].
7. Branches A+B converge → **dysmegakaryopoiesis**: expansion of megakaryocyte progenitors (MkP) with loss of polyploid/terminal MKs and reduced platelet output (demonstrated) [PMID: 41538704].
8. Reduced effective platelet production → **mild-to-moderate thrombocytopenia with normal-sized platelets** and compensatory elevated serum thrombopoietin (clinical observation) [PMID: 21467542].
9. **Separate long-term branch — leukemogenesis:** the germline gain-of-expression background + acquisition of **somatic driver mutations** in hematopoietic regulators → clonal evolution → **MDS / AML / CMML** (demonstrated/inferred two-hit model) [PMID: 35751752].

Mechanism schematic



Supporting quotes. "persistent ANKRD26 expression in isolated MKs increased signaling via the thrombopoietin/myeloproliferative leukemia virus oncogene (MPL) pathway and impaired proplatelet formation by MKs. Importantly, we demonstrated that ERK inhibition completely rescued the in vitro proplatelet formation defect" [PMID: 24430186]. "ANKRD26 interacts with and crucially modulates the activity of MPL, EPOR and G-CSFR, three homodimeric type I cytokine receptors that regulate blood cell production. Higher than normal levels of ANKRD26 prevent the receptor internalization that leads to increased signaling and cytokine hypersensitivity" [PMID: 36794499]. "Confocal imaging localized ANKRD26 to the centrosome, implicating it in mitotic regulation during MK maturation. Mechanistically, we discovered that elevated ANKRD26 induces apoptosis in polyploid MKs via JUNB-mediated transcriptional activation" [PMID: 41538704].

Pathways and ontology mapping. Molecular pathways: TPO–MPL signaling; MAPK/ERK cascade (KEGG hsa04010); type I cytokine receptor signaling (MPL, EPOR, CSF3R). Cellular processes: megakaryocyte maturation/polyploidization (endomitosis), proplatelet formation, apoptosis. Protein dysfunction: ANKRD26 overexpression is a **gain-of-function at the expression level** that dysregulates receptor trafficking. **Suggested GO terms:** GO:0030220 (platelet formation), GO:0002576 (platelet degranulation), GO:0000278 (mitotic cell cycle), GO:0006915 (apoptotic process), GO:0019221 (cytokine-mediated signaling pathway). **Suggested CL terms:** CL:0000556 (megakaryocyte), CL:0000763 (myeloid cell), CL:0000037 (hematopoietic stem cell), CL:0000233 (platelet).

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7. Anatomical Structures Affected

- **Primary organ/tissue:** Bone marrow (UBERON:0002371), specifically the megakaryocytic lineage; peripheral blood platelets.
- **Body system:** Hematopoietic / hematologic system (UBERON:0002390 bone marrow; UBERON:0000178 blood).
- **Cell populations:** Megakaryocytes and their progenitors (CL:0000556 megakaryocyte; megakaryocyte progenitor), hematopoietic stem/progenitor cells (CL:0000037), platelets (CL:0000233). Multilineage receptor modulation (MPL/EPOR/G-CSFR) means erythroid and granulocytic lineages can be subtly affected (occasional high hemoglobin/leukocytes) [PMID: 21467542, 36794499].
- **Subcellular level: Centrosome** (GO:0005813) — site of ANKRD26 localization implicating mitotic regulation; cytoplasmic membrane receptor-trafficking machinery; nucleus (JUNB/p21 transcriptional program).
- **Localization/lateralization:** Systemic (bone marrow throughout the skeleton); not lateralized. *Not applicable* for unilateral/bilateral descriptors.

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8. Temporal Development

- **Onset:** Congenital/lifelong thrombocytopenia (present from birth, often detected incidentally in childhood or adulthood). Onset is **insidious/chronic**, not acute.

- **Progression of the thrombocytopenia:** Generally **stable** and non-progressive over life; the platelet count does not typically deteriorate from the primary lesion alone.
- **Critical transition:** The clinically decisive event is **malignant transformation** to MDS/AML/CMML, which can occur at variable ages in adulthood and reflects acquisition of somatic driver mutations.
- **Diagnostic latency:** Genetic diagnosis is typically **delayed ~20 years** from first detection of thrombocytopenia PMID: 40670159.
- **Duration:** Chronic, lifelong. No spontaneous remission of the thrombocytopenia.

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9. Inheritance and Population

- **Inheritance: Autosomal dominant** (germline). Offspring of an affected parent have a 50% transmission risk.
- **Penetrance:** High penetrance for the thrombocytopenia; **incomplete and variable penetrance** for the malignancy phenotype.
- **Expressivity:** Variable (bleeding, blood-count abnormalities, malignancy risk differ among carriers).
- **Prevalence/frequency:** THC2 is rare at the population level but **relatively frequent among inherited thrombocytopenias** — it accounted for 21/210 (~10%) families in the Italian inherited-thrombocytopenia database: "Because THC2 affected 21 of the 210 families in our database, it has to be considered one of the less rare forms of inherited thrombocytopenia" PMID: 21467542. It is systematically **under-diagnosed** because inherited thrombocytopenias are frequently missed — "we suspect that inherited thrombocytopenias are underreported because their diagnosis is often missed" PMID: 12745278.
- **Anticipation / mosaicism / founder effects / consanguinity:** No genetic anticipation (not a repeat-expansion disorder). As a dominant disorder, consanguinity is not a driver. No established founder effect, though recurrent hotspot mutations arise independently. *Specific carrier-frequency and founder data: not available.*
- **Sex ratio / ethnicity / geography:** No strong sex bias reported; described across European (Italian) and Japanese registries, consistent with broad geographic distribution. Precise sex ratios and population-specific prevalence are *not available*.

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10. Diagnostics

Clinical/laboratory workup. - **CBC with platelet indices:** Mild-to-moderate thrombocytopenia with **normal mean platelet volume (MPV)** — the key discriminator. Platelet size/volume is the primary branch point in inherited-thrombocytopenia diagnostic algorithms: "The different forms have been classified into 3 groups depending on platelet volume … it is one of the most useful [criteria] in diagnostic algorithms" PMID: 12161364. ANKRD26-RT falls in the **normal-sized-platelet group** PMID: 36591709. - **Peripheral blood smear:** Normal-sized platelets; useful to exclude macrothrombocytopenias. - **Platelet function testing:** Historically normal aggregation; more recent data show frequently impaired aggregation and heterogeneous granule secretion; frequent GPIa (ITGA2) and α-granule deficiency [PMID: 21467542, 40670159]. - **Serum thrombopoietin:** Often elevated, consistent with a production (dysmegakaryopoietic) defect [PMID: 21467542]. - **Bone marrow:** Dysmegakaryopoiesis; single-cell studies show MkP expansion and polyploid-MK reduction [PMID: 41538704].

Genetic testing (definitive). Diagnosis rests on **targeted sequencing of the *ANKRD26* 5′UTR**, performed either as single-gene testing or within NGS **inherited-platelet-disorder / MDS-AML-predisposition panels** that also cover *RUNX1* and *ETV6*. WES/WGS can detect the variant but the non-coding 5′UTR hotspot must be specifically covered and interpreted. Karyotyping/CMA/FISH are reserved for assessing clonal evolution, not for primary diagnosis.

Clinical criteria & differential diagnosis. Key differentials are the other **normal-platelet-size inherited thrombocytopenias with malignancy risk — RUNX1-RT and ETV6-RT** — and, critically, **acquired immune thrombocytopenia (ITP)**, which is the most common and most consequential misdiagnosis (leading to inappropriate immunosuppression or splenectomy).

Screening. No newborn screening. **Cascade genetic testing** of at-risk relatives is recommended once the familial 5′UTR variant is identified.

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11. Outcome / Prognosis

- **Thrombocytopenia prognosis:** Generally benign; bleeding is usually mild and life expectancy from the platelet disorder alone is normal.
- **Dominant prognostic driver — malignancy:** Inherited thrombocytopenia with germline *RUNX1*, *ETV6* or *ANKRD26* variants "carries a high risk (10%-45%) of developing

haematological malignancy (IT-HM)" PMID: 40670159. Reported ANKRD26-associated neoplasms include **AML, MDS, and CMML** PMID: 35751752; in the 2025 cohort, ANKRD26-RT cases included B-ALL (Ph+), MDS and breast cancer, with one-third of the combined IT cohort developing malignancy [PMID: 40670159].

- **Transformation mechanism:** "Acute myeloid leukemia, myelodysplastic syndrome, and chronic myelomonocytic leukemia have been reported to occur in the context of germline variants in ANKRD26, with the development of somatic driver mutations in hematopoietic regulators playing an important role in malignant transformation" (two-hit model) [PMID: 35751752].
- **Prognostic factors:** Development of cytopenias beyond baseline, new dysplasia, or clonal cytogenetic/molecular abnormalities signal progression and warrant urgent evaluation.
- **QoL/disability:** Low from the thrombocytopenia; substantial if malignancy develops.

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12. Treatment

General principle: No disease-modifying therapy exists. Management is **supportive**, and most patients with mild bleeding require **no chronic treatment**.

Pharmacotherapy / TPO-receptor agonists. The TPO-RA **eltrombopag** (NCIT:C62492) effectively raises platelet counts perioperatively. In a case series of inherited thrombocytopenias that included ANKRD26-RT, eltrombopag at 50 mg/day started ~3 weeks before surgery produced a favorable platelet response in **14 of 15 procedures**, preventing bleeding complications and offering an alternative to platelet transfusion: "Nine patients with confirmed ITs (MYH9-related disease, ANKRD26-related thrombocytopenia [-RT], ITGA2B/ITGB3-RT, ACTN1-RT) underwent 15 surgical procedures. Eltrombopag was administered at 50 mg/d, initiated approximately 3 weeks prior to surgery" and "A favorable platelet response occurred in 14 of 15 procedures" PMID: 42620598.

Supportive/interventional. - Perioperative platelet support (platelet transfusion or TPO-RA) for surgery/trauma. - **Avoid unnecessary antiplatelet drugs** and avoid splenectomy / immunosuppression driven by ITP misdiagnosis. - Antifibrinolytics (e.g., tranexamic acid) for mucosal bleeding where appropriate (general IT practice).

Advanced therapeutics. Allogeneic hematopoietic stem cell transplantation (HSCT; NCIT:C15431) is reserved for progression to MDS/AML. Germline status critically informs **donor selection — related donors must be screened to avoid using a donor carrying the**

familial *ANKRD26* variant: the role of HSCT "in managing MDS, particularly in patients with germline syndromes, is reviewed, emphasizing the need for optimal donor selection and personalized treatment approaches" PMID: 39443230. No approved gene, RNA, or targeted therapy exists for THC2 itself; ERK-pathway inhibition rescues the proplatelet defect in vitro and is a mechanistic lead, not a clinical therapy [PMID: 24430186].

Pharmacogenomics / personalized medicine: Treatment is genotype-aware chiefly through germline-status-guided surveillance and transplant donor selection rather than drug-metabolism pharmacogenomics.

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13. Prevention

- **Primary prevention: None** — THC2 is a germline autosomal-dominant condition; the disease cannot be prevented in a mutation carrier.
- **Secondary prevention: Periodic CBC and bone-marrow surveillance** for early detection of clonal evolution/MDS-AML; early recognition of dysplasia enables timely intervention.
- **Tertiary prevention:** Avoid unnecessary antiplatelet drugs, immunosuppression, and splenectomy from ITP misdiagnosis; provide perioperative platelet support; treat malignant transformation promptly.
- **Genetic counseling:** Central to management — 50% offspring transmission risk, cascade testing of relatives, and **informed donor selection for HSCT**. *ANKRD26* is confirmed as a germline MDS-predisposition gene: "hereditary conditions that affect platelet production and count, such as RUNX1, ETV6, and ANKRD26 … contribute to an increased risk of MDS" PMID: 39443230.
- **Reproductive options:** Preimplantation genetic testing (PGT) and prenatal diagnosis are available once the familial 5′UTR variant is known.

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14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *ANKRD26* is conserved in mouse (*Ankrd26*; NCBI Taxon:10090). *ANKRD26* is the ancestor of the primate-specific *POTE* gene family [PMID: 21211618].

- **Natural disease in other species:** No naturally occurring animal homolog of THC2 (gain-of-expression thrombocytopenia) has been described. *Not applicable* for companion-animal/wildlife natural disease.
- **Comparative biology:** Importantly, the mouse *Ankrd26* **knockout models loss of function, not THC2.** *Ankrd26*−/− mice develop hyperphagia, obesity and gigantism via a **ciliopathy** of hypothalamic appetite/energy-homeostasis centers — and do **not** develop thrombocytopenia: "disruption of this gene causes hyperphagia, obesity and gigantism in mice. In the present study, we looked for the mechanisms that induce hyperphagia in the Ankrd26-/- mice and found defects in primary cilia in regions of the central nervous system that control appetite and energy homeostasis" PMID: 24633808. This divergence is itself evidence that THC2 is a gain-of-expression disease, not haploinsufficiency.
- **Zoonotic potential:** None (non-transmissible genetic disorder).

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15. Model Organisms

Model	Type	What it captures	Limitation
Ankrd26−/− mouse	Mammalian knockout (loss of function)	ANKRD26 LoF biology (ciliopathy, obesity)	**Does NOT recapitulate THC2**; no thrombocytopenia [PMID: 24633808]
Patient primary megakaryocytes	Human ex vivo	RUNX1/FLI1-binding loss, MPL/ERK hyperactivation, proplatelet defect, ERK-rescue	Limited cell numbers; culture artifacts [PMID: 24430186]
Patient bone marrow single-cell (scRNA-seq)	Human ex vivo	MkP expansion, polyploid-MK apoptosis, JUNB-p21 mechanism	Observational; 4 patients [PMID: 41538704]
Patient-derived iPSC line (e.g., CGOi001-A)	Human in vitro	ANKRD26-mutant hematopoietic differentiation	Differentiation efficiency; emerging resource (P 41643484)
Cytokine-receptor overexpression cell lines	In vitro	MPL/EPOR/G-CSFR trafficking/signaling	Reductionist [PMID: 36794499]

Key point: Because THC2 is caused by a **gain of expression**, the standard mouse knockout models the wrong allele direction. The faithful disease-relevant systems are **human patient-derived** (primary MKs, iPSC-derived hematopoietic cells, single-cell bone marrow profiling). No fully faithful animal model of the gain-of-expression thrombocytopenia currently exists — a notable knowledge gap.

Mechanistic Model / Interpretation

The pathophysiology of THC2 is now coherent and converges from two independent experimental programs onto a unified model. The **initiating lesion is regulatory, not coding**: 5′UTR mutations in a conserved 19-bp window strip away RUNX1/FLI1 repression, so *ANKRD26* — normally switched off as megakaryocytes mature — stays on. The downstream consequences bifurcate but reinforce the same endpoint:

- A **signaling branch** (Bluteau et al.; Basso-Valentina et al.) in which excess ANKRD26 protein stabilizes MPL/EPOR/G-CSFR, raises TPO/MPL signaling, and hyperactivates MAPK/ERK, blocking proplatelet formation — causally proven because ERK inhibition fully rescues proplatelet formation.
- An **apoptotic branch** (Chen et al., single-cell) in which centrosome-localized ANKRD26 drives JUNB→p21 transcription to kill polyploid megakaryocytes, depleting the mature cells that make platelets while progenitors (MkP) accumulate.

Both branches yield **dysmegakaryopoiesis** → fewer platelets of normal size → the characteristic normocytic, mild-to-moderate thrombocytopenia with compensatory high TPO. Layered on top, a **leukemogenic trajectory** emerges over decades as somatic driver mutations accrue on the germline background, explaining the 10–45% malignancy predisposition that dominates prognosis. The *GATA2*-positive / *RUNX1-FLI1*-negative regulatory tug-of-war over *ANKRD26* ties the predisposition genes into a single regulatory circuit.

Clinically, the single most actionable insight is **diagnostic discrimination**: normal platelet size + autosomal-dominant family history should trigger targeted *ANKRD26* 5′UTR sequencing and divert the patient away from an ITP label, averting harmful immunosuppression/splenectomy and enrolling them in lifelong malignancy surveillance.

Evidence Base

PMID	Role	Key contribution
21211618	Landmark (genetics)	Identifies *ANKRD26* 5′UTR 19-bp hotspot; gain-of-expression, not haploinsufficiency
24430186	Landmark (mechanism)	RUNX1/FLI1-binding loss → MPL/MAPK-ERK hyperactivation → proplatelet defect; ERK-inhibition rescue
36794499	Mechanism	ANKRD26 modulates MPL/EPOR/G-CSFR internalization → cytokine hypersensitivity
41538704 / 42911644	Recent landmark	Single-cell: MkP expansion, polyploid-MK apoptosis via centrosomal ANKRD26 + JUNB-p21
21467542	Landmark (clinical)	78 patients/21 families: normocytic platelets, mild bleeding, GPIa/α-granule deficiency, leukemia risk; ~10% of IT families
40670159	Recent cohort	10–45% malignancy risk; bleeding in 25–30%; impaired aggregation in 90%; ~20-yr diagnostic delay
35751752	Review	AML/MDS/CMML spectrum; somatic-driver two-hit transformation
40954090	Registry	Japanese registry: novel variant disrupts RUNX1 site → overexpression, reduced proplatelets, lower ploidy
38817334	Mechanism/ modifier	GATA2 as positive regulator of *ANKRD26*; confirms core THC2 definition
24633808	Model organism	*Ankrd26* KO = ciliopathy/obesity, NOT thrombocytopenia (supports non-haploinsufficiency)
42620598	Treatment	Eltrombopag effective perioperatively (14/15 procedures) incl. ANKRD26-RT
39443230	Management	Germline-informed donor selection/counseling for HSCT in MDS predisposition
12161364 / 12745278 / 36591709	Diagnostics	Platelet-size classification; under-diagnosis of inherited thrombocytopenia

Evidence sources span **human clinical cohorts/registries** (21467542, 40670159, 40954090), **human ex vivo/in vitro functional studies** (24430186, 36794499, 41538704), and **model organism** work (24633808). No contradictions exist in the mechanistic model; the main evolution is the refinement of "normal platelet aggregation" (2011) toward "frequently impaired/heterogeneous function" (2025).

Limitations and Knowledge Gaps

1. **No faithful animal model** of the gain-of-expression disease; the mouse knockout models the opposite allele direction.
2. **Malignancy risk is a group estimate** (10–45% across RUNX1/ETV6/ANKRD26) rather than a precise *ANKRD26*-specific, age-stratified figure; penetrance determinants are incompletely defined.
3. **Variant interpretation** for novel non-canonical 5′UTR variants and coding VUS remains challenging; functional assays (reporter/transcript quantitation) are not universally standardized.
4. **Epidemiology** — precise prevalence/incidence, sex ratio, and ethnicity-specific data are not well quantified; the disorder is under-diagnosed.
5. **No QoL instruments** validated specifically for THC2.
6. **Surveillance protocols** (frequency/modality of CBC and marrow monitoring) are not standardized by prospective evidence; preventive/leukemia-interception therapies are aspirational.

Proposed Follow-up Experiments / Actions

1. **Build a conditional gain-of-expression model** (megakaryocyte-specific *Ankrd26* overexpression or 5′UTR knock-in of a RUNX1-site-disrupting mutation) to recapitulate normocytic thrombocytopenia and test leukemia predisposition in vivo.
2. **iPSC-based mechanistic dissection** using patient lines (e.g., CGOi001-A) to validate the JUNB-p21 apoptotic branch and screen ERK-pathway or JUNB/p21 modulators as proplatelet-restoring agents.

3. **Prospective natural-history registry** to derive *ANKRD26*-specific, age-stratified malignancy penetrance and define evidence-based surveillance intervals.
4. **Standardized diagnostic pathway**: embed MPV + autosomal-dominant history as a trigger for *ANKRD26* 5′UTR sequencing in thrombocytopenia panels to reduce ITP misdiagnosis and the ~20-year diagnostic delay.
5. **Functional classification pipeline** (luciferase reporter + transcript quantitation + RUNX1/FLI1 ChIP) to resolve non-canonical 5′UTR VUS for ACMG/AMP classification.
6. **Clonal-surveillance biomarkers**: longitudinal somatic-mutation tracking (serial NGS/liquid biopsy of hematopoietic clones) to detect pre-leukemic evolution and enable early intervention.

Report compiled from a 5-iteration autonomous investigation; 10 confirmed findings, 29 papers reviewed. Evidence types: human clinical cohort/registry, human ex vivo/in vitro functional, and model organism.
